PROBLEM 50

Double Pendulum

Original statement · Problems.pdf, p. 35

Kinematic Linkage

A double pendulum consists of two heavy point masses P and Q, each of mass m, connected to the origin O and to each other by two rigid, massless bars of length ℓ. We define the inertial frame S_0 with origin O and vector basis $\mathcal{B}_0$. We define the two angles θ and ϕ to describe the configuration of the system.

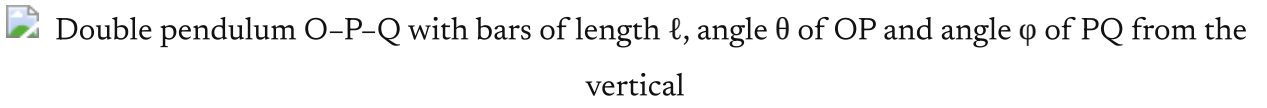


Figure: Double pendulum O–P–Q with bars of length ℓ, angle θ of OP and angle φ of PQ from the vertical (Problems.pdf, p. 35).

(a) Write down the general expressions of the position, velocity, and acceleration vectors of P and Q, using θ, ϕ and their derivatives.

(b) Write down all forces acting on P and Q, introducing new unknowns as needed.

(c) Write down the equations of motion for θ and ϕ, free of other unknowns.

(d) Find and write down one conservation law of this problem.

(e) If the initial conditions are $\theta = \phi = \pi/4, \dot{\theta} = \dot{\phi} = 0$, will Q ever cross the Ox axis? Justify your answer.

(f) Consider now that we impose the additional constraint $\phi = 2\theta$. Compute the natural frequency of oscillation of the system under the assumption of small angles.

(Hint: The mechanical energy of an ideal harmonic oscillator of natural frequency ω_0 and mass m can be written as $\frac{1}{2}m(\dot{x}^2 + \omega_0^2 x^2)$).

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two point masses P and Q, each of mass m, connected by two rigid massless bars of length ℓ.
- First bar connects origin O to P, making angle θ with downward vertical.
- Second bar connects P to Q, making angle ϕ with downward vertical.
- **CDOFs:** 2 degrees of freedom: (θ, ϕ).
- **Parameters:** $m, \ell, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Coordinate axes: Ox horizontal to the right, Oz vertically upwards (downward vertical is $-\mathbf{k}$).
- Position of P:

$$\mathbf{r}_P = \ell \sin\theta\, \mathbf{i} - \ell \cos\theta\, \mathbf{k}$$

- Position of Q:

$$\mathbf{r}_Q = \mathbf{r}_P + \ell \sin\phi\, \mathbf{i} - \ell \cos\phi\, \mathbf{k} = \ell(\sin\theta + \sin\phi)\, \mathbf{i} - \ell(\cos\theta + \cos\phi)\, \mathbf{k}$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Velocity and Acceleration Vectors

1. Velocity of P:

$$\mathbf{v}_P = \ell\dot{\theta}\cos\theta\,\mathbf{i} + \ell\dot{\theta}\sin\theta\,\mathbf{k}$$

$$v_P^2 = \ell^2\dot{\theta}^2$$

2. Velocity of Q:

$$\mathbf{v}_Q = \ell(\dot{\theta}\cos\theta + \dot{\phi}\cos\phi)\,\mathbf{i} + \ell(\dot{\theta}\sin\theta + \dot{\phi}\sin\phi)\,\mathbf{k}$$

$$v_Q^2 = \ell^2\left[\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}(\cos\theta\cos\phi + \sin\theta\sin\phi)\right] = \ell^2\left[\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos(\theta-\phi)\right]$$

Part (b)-(c): Equations of Motion

Total Kinetic Energy:

$$T = \frac{1}{2}mv_P^2 + \frac{1}{2}mv_Q^2 = \frac{1}{2}m\ell^2\left[2\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos(\theta-\phi)\right]$$

Total Potential Energy (taking datum at $z=0$):

$$V = mgz_P + mgz_Q = -mg\ell\cos\theta - mg\ell(\cos\theta + \cos\phi) = -mg\ell(2\cos\theta + \cos\phi)$$

Lagrange's equations $\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial T}{\partial q} = -\frac{\partial V}{\partial q}$:

1. For θ:

$$2\ell\ddot{\theta} + \ell\ddot{\phi}\cos(\theta - \phi) + \ell\dot{\phi}^2\sin(\theta - \phi) + 2g\sin\theta = 0$$

2. For ϕ:

$$\ell\ddot{\phi} + \ell\ddot{\theta}\cos(\theta - \phi) - \ell\dot{\theta}^2\sin(\theta - \phi) + g\sin\phi = 0$$

Part (d): Conservation Law

The system is conservative with scleronomic constraints. Total mechanical energy is constant:

$$E = \frac{1}{2}m\ell^2\left[2\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos(\theta - \phi)\right] - mg\ell(2\cos\theta + \cos\phi) = \text{constant}$$

Part (e): Will Q Ever Cross the Ox Axis?

- Given initial state: $\theta(0) = \phi(0) = \pi/4$, released from rest ($\dot{\theta}(0) = \dot{\phi}(0) = 0$).
- Initial energy:

$$E_0 = -mg\ell\left(2\cos\frac{\pi}{4} + \cos\frac{\pi}{4}\right) = -3mg\ell\frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2}mg\ell \approx -2.121\,mg\ell$$

- For Q to cross the Ox axis, its vertical coordinate z_Q must reach 0:

$$z_Q = -\ell(\cos\theta + \cos\phi) = 0 \implies \cos\theta + \cos\phi = 0$$

At that instant, the minimum potential energy of the system is:

$$V = -mg\ell(2\cos\theta + \cos\phi) = -mg\ell(\cos\theta + (\cos\theta + \cos\phi)) = -mg\ell\cos\theta$$

Since $\cos\theta \leq 1, V \geq -mg\ell$.

The required energy to cross the axis would be:

$$E_{req} \geq V \geq -mg\ell$$

However, the initial energy is $E_0 = -2.121\, mg\ell < -mg\ell$.

Because $E_0 < E_{req}$, the kinetic energy would have to be negative, which is physically impossible!

Conclusion: Q can **never** cross the Ox axis.

Part (f): Imposed Constraint $\phi = 2\theta$ and Small-Angle Natural Frequency

When $\phi = 2\theta, \dot{\phi} = 2\dot{\theta}$.

For small angles $\theta \ll 1$:

$$\cos(\theta - \phi) = \cos(-\theta) = \cos\theta \approx 1$$

Kinetic energy:

$$T = \frac{1}{2}m\ell^2\left[2\dot{\theta}^2 + (2\dot{\theta})^2 + 2\dot{\theta}(2\dot{\theta})(1)\right] = \frac{1}{2}m\ell^2[2+4+4]\dot{\theta}^2 = 5m\ell^2\dot{\theta}^2$$

Potential energy with $\cos u \approx 1 - \frac{u^2}{2}$:

$$V = -mg\ell(2\cos\theta + \cos(2\theta)) \approx -mg\ell\left[2\left(1 - \frac{\theta^2}{2}\right) + \left(1 - \frac{4\theta^2}{2}\right)\right]$$

$$V \approx -mg\ell[3 - 3\theta^2] = \text{const} + 3mg\ell\theta^2$$

Total energy of the equivalent oscillator:

$$E = 5m\ell^2\dot{\theta}^2 + 3mg\ell\theta^2 = \text{constant}$$

Differentiating with respect to time:

$$10m\ell^2\dot{\theta}\ddot{\theta} + 6mg\ell\theta\dot{\theta} = 0$$

Divide by $10m\ell^2\dot{\theta}$:

$$\ddot{\theta} + \frac{6g}{10\ell}\theta = 0 \implies \ddot{\theta} + \frac{3g}{5\ell}\theta = 0$$

The natural frequency of oscillation is:

$$\omega_0 = \sqrt{\frac{3g}{5\ell}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- The mechanical coupling $\phi = 2\theta$ acts as an internal kinematic constraint that stiffens the double pendulum into an equivalent single degree of freedom harmonic oscillator with effective mass $10m$ and effective restoring stiffness $6mg/\ell$, yielding the frequency $\omega_0 = \sqrt{3g/(5\ell)}$.

- **Dimensional check:** $[T] = [m\ell^2\dot\theta^2] = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-2} = \mathrm{J}$, $[V] = [mg\ell] = \mathrm{J}$, and $[\omega_0] = \sqrt{[g]/[\ell]} = \mathrm{s}^{-1}$ (the effective inertia $10m\ell^2$ and stiffness $6mg\ell$ are expressed in the coordinate θ; with $x = \ell\theta$ they become $10m$ and $6mg/\ell$).
- **Numerical check of part (e) (RK4, step 10^{-3}, $g = 9.81\ \mathrm{m/s^2}$, $m = \ell = 1$, 60 s from $\theta = \phi = \pi/4$ at rest):** the full nonlinear equations of part (c) give a maximum of $z_Q/\ell = -1.3375$, always negative, so Q never reaches the Ox axis, consistent with the energy bound $V \geq -mg\ell$ against $E_0 = -2.121\,mg\ell$.
- **Numerical check of part (f):** the exact one-degree-of-freedom Lagrangian with $\phi = 2\theta$ (without the small-angle approximation) integrated by RK4 from $\theta = 10^{-3}$ rad has half-period 1.2950 s, against $\pi/\omega_0 = \pi\sqrt{5\ell/(3g)} = 1.2949$ s.

Source: Problems.pdf (Page 35)